CHAPTER 01 / ANATOMY

Understand your body

Know what moves you.

Explore the muscles behind every movement.

How to use this guide

Read the explanations, try the practical examples and complete the worksheet. All training logs and menus are illustrative. Use the website for interactive tools; this document remains useful offline.

01	02	03
Understand	Put into practice	Review

Scope and safety

General education for adults, not a medical diagnosis, personalized treatment or a guaranteed transformation. Stop exercise for sharp pain or unusual symptoms. A qualified clinician, dietitian or coach can help when individual circumstances require assessment.

What you will learn

01. A system of levers

02. Train movements, understand muscles

03. Your lower-body engine

04. Use the model thoughtfully

01 / A system of levers

Bones provide structure. Joints allow motion. Skeletal muscles pull on tendons to move or stabilize those joints. A muscle can produce force while shortening, lengthening or holding position. These are concentric, eccentric and isometric actions.

PUT IT INTO PRACTICE: In a curl, lifting, lowering and pausing all involve muscle force.

02 / Train movements, understand muscles

Chest muscles bring the upper arm across the body. The back contains several muscles that pull or stabilize the shoulder region. Deltoids help raise the arm. Biceps bend the elbow; triceps straighten it. Forearm muscles move and stabilize the wrist and fingers.

PUT IT INTO PRACTICE: Presses, rows and pulls involve several muscles at once.

03 / Your lower-body engine

Quadriceps extend the knee. Hamstrings bend the knee, and most also extend the hip. Glutes extend or stabilize the hip. Calves point the foot downward. Abdominal muscles move and brace the trunk. Joint positions change how a movement distributes work.

PUT IT INTO PRACTICE: A squat trains more than the front of the thigh; the hips and trunk contribute too.

04 / Use the model thoughtfully

Explore front and back views, then select a region. The model is a simplified map, not a diagnostic or surgical anatomy model. Bodies vary in shape. You do not need to feel a dramatic burn in every muscle for a set to be useful.

PUT IT INTO PRACTICE: Use a comfortable range and ask a qualified coach to check unfamiliar lifts.

Practical reference

Muscle group	Location & action	Exercise examples
Chest	Front of upper torso. Brings the upper arm across the body; assists shoulder flexion.	Push-up, dumbbell press
Back	Back and sides of torso. Pulls the upper arm down and back; upper-back muscles help position the shoulder blades.	Lat pulldown, cable row
Shoulders	Around the shoulder joint. Raises the upper arm; different portions assist different directions.	Shoulder press, lateral raise
Biceps	Front of upper arm. Bends the elbow and turns the palm upward.	Dumbbell curl
Triceps	Back of upper arm. Straightens the elbow.	Triceps pushdown
Forearms	Between elbow and wrist. Moves and stabilizes the wrist and fingers.	Farmer carry, wrist curl
Quadriceps	Front of thigh. Extends the knee; rectus femoris also assists hip flexion.	Goblet squat, leg extension
Hamstrings	Back of thigh. Bends the knee; most of the group also extends the hip.	Romanian deadlift, leg curl
Glutes	Back and side of hip. Extends and stabilizes the hip.	Hip thrust, split squat
Calves	Back of lower leg. Points the foot downward at the ankle.	Calf raise
Abdominals	Front and sides of trunk. Moves and braces the trunk.	Dead bug, plank

Your study worksheet

Knowledge check

Which muscle group straightens the knee?

A. Hamstrings

B. Quadriceps

C. Biceps

Reflect and apply

Prompt	Your response
One idea I can explain in my own words	
One action I can try this week	
What I will observe or measure	
What would make me seek professional guidance	
My next review date	

Chapter summary

Muscles work together across joints. Understand the action before choosing the exercise.

Answer and explanation

B. The quadriceps are knee extensors.

Sources & limitations

These sources inform the educational principles. Original examples and planning templates are not study findings. No institution listed below endorses this journal. References checked November 2026.

OpenStax — Anatomy and Physiology 2e

https://openstax.org/books/anatomy-and-physiology-2e/pages/11-introduction

Interpretation matters

Research describes groups under particular conditions. Individual responses vary. The website models are simplified illustrations, not precise physiological simulations. Food values and calorie estimates are approximate. Use consistent observation and individualized professional support when needed.